PRACTICAL EXERCISES

Practical 1 Pressure Testing

1. Testing of a pressure transducer.
2. Ranging and Calibration of a 4-20 mA pressure transmitter.
3. Investigation of positive and negative pressures.

Practical 2 Level measurement with Ultrasonics

1. Tank level measurement with ultrasonics.
2. Calibration of 4-20mA signal for process measurement.
3. Functionality of ultrasonics with varying materials and surfaces.

Practical 3 Temperature

1. Temperature response and accuracy.
2. Thermocouple grounded, ungrounded, sheathed.
3. RTD, 2/3/4 wire comparison and thermowell response.
4. Thermocouple, Extension leads compared with compensation leads.

Practical 4 Simulation and Temperature Transmitter Calibration

1. Transmitter simulation.
2. Zero and span adjustment with 4-20mA signal.
3. Transmitter calibration - accuracy and error calculations.

Practical 5 Valve sizing (software)

1. Valve selection and sizing using typical software.

Notes to the Practicals

The following notes should be read in conjunction with the extracts from the relevant manuals supplied with the equipment. If in doubt with any procedure described below consult the instructor. He is there to assist you to eliminate any safety problems and to protect the equipment from any damage!

Some attendees will be very familiar with some items of equipment and procedures. Others may not be.

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Practical Exercise 1 Pressure Testing

1.1 Objectives

- To check, range and calibrate various types of indicators and 4-20mA pressure transducers.

1.2 Equipment Required

- Pressure calibrator (Druck DPI-610)
- Mechanical IMT Bourdon tube type indicator **PI 101**
- Endress and Hauser PMC 133 Pressure Transducer **PT 102**
- Rosemount model 3051 DP pressure transmitter **PT 103**
- Connection cables and pressure tubing

1.3 Procedure - Bourdon Tube pressure indicator PI 101

Check equipment details:

Tag no.: PI 101
Manufacturer: IMT
Range: 0-400kPa (0-59psig)

Connect the Bourdon pressure indicator to the pressure calibrator test port with the ¼" tube supplied.

Switch on the calibrator and close the pressure-release valve.

Set the pressure calibrator for kPa as follows:
- Press the '**SETUP**' button.
- Cursor down to '**SETTINGS**' and press the '**Return Arrow**' key.
- Select '**UNITS**'
- Use '**F1**' to remove the MPa selection
- Cursor **up once** and select kPa with the '**F1**' key.
- Press the '**Return Arrow**' to return to the settings menu.
- Press '**EXIT**' to return to the setup menu.
- Press '**EXIT**' to return to normal operation.

Configure the Output window to display kPa.
- Press the '**OUTPUT**' key.
- Press '**F2**' for units.
- Press the '**Return Arrow**' to select kPa.

Confirm that the pump action is in the pressure mode (Rotate the base of the hand pump in the clockwise direction). Pump up the pressure until the indicator reads full scale (400kPa), and note the reading on the calibrator.

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Determine the difference between the indicator and the calibrator and calculate the inaccuracy as a percent of the meter span.

Calibrator pressure (kPa)	PI 101 pressure (kPa)	Error (%)
0		
200		
400		

1.4 Procedure - PMC 133 pressure transmitter PT 102

Check equipment details:

Tag no.: PT 102
Manufacturer: Endress + Hauser
Model: PMC 133
Serial: 1R1F2A6F16
Range: 0-1000kPa

Connect pressure tapping's between the Endress and Hauser PMC 133 transmitter and the pressure calibrator test port.

Make the electrical connections between the transmitter and the calibrator – connecting the pressure transmitter terminals 1(+) and 2(-) to the "Xmtr" terminals on the Pressure Calibrator (see attached diagram).

Press "OUTPUT" button and use scroll arrows to scroll down to "24 VOLT OFF". Press left arrow key to display "24 VOLT ON". Press enter and ensure that "24 VOLT ON" is displayed on "OUTPUT" panel of display.

Open release valve and ensure that pressure reads 0.0000 kPa. If not, press the "ZERO" button and than "F2". At the next prompt press "F1".

Adjust the transmitter "Z" control until 4 mA is obtained on "CURRENT" panel of display.

Close release valve and, using hand pump and vernier adjustment, set the pressure to 250 kPa. Adjust the transmitter "S" control until 20mA is obtained in the top display.

Set the pressure for 0 kPa and recheck the transmitter output is 4mA. Adjust if required.

Check and record milliamp values of output for each 10% of pressure span. Plot these values to confirm linearity of the transducer and transmitter.

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Set Pressure (kPa)	Required current (mA)	Measured current (mA)
0	4	
25	5.6	
50	7.2	
75	8.8	
100	10.4	
125	12.0	
150	13.6	
175	15.2	
200	16.8	
225	18.4	
250	20.0	

Refer to the Manufacturers brochure for the following:

What is the sensing device used? (Ref page 3)

What types of pressure can this device measure?
- Gauge
- Absolute
- Differential

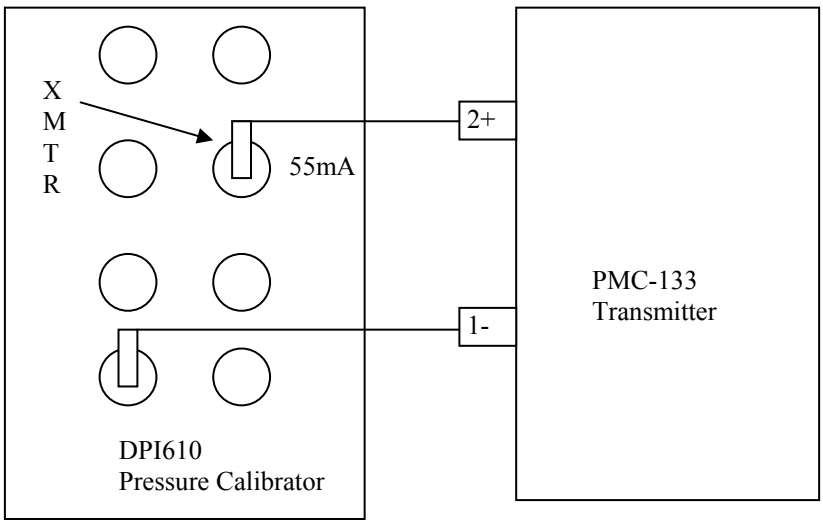


Figure E1.1

1.5 Procedure - Rosemount Pressure transmitter PT 103

Check equipment details:

Tag no.: PT 103
Manufacturer: Rosemount

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Model: 3051C
Range: 0-50kPa

IMPORTANT: Locate and identify the H and L on the base of the transmitter. The H indicates the high-pressure side and the L indicates the low pressure tapping side.

PART 1.

Connect the pressure calibrator to the **HIGH** side of the Rosemount pressure transmitter using ¼" pipe.

Make the electrical connections between the transmitter and the calibrator – connecting the pressure transmitter terminals (+) and (-) to the "Xmtr" terminals on the Pressure Calibrator (see attached diagram).

Press "OUTPUT" button and use scroll arrows to scroll down to "24 VOLT OFF". Press left arrow key to display "24 VOLT ON". Press "Enter" and ensure that "24 VOLT ON" is displayed on "OUTPUT" panel of display.

Open the release valve and ensure that the Pressure display reads 0.0000 kPa. If not, press "ZERO" button and then "F2". At the next prompt press "F1".

Adjust the transmitter zero by releasing the "Z" switch only until 4 mA is obtained on "CURRENT" panel of display. **DO NOT** touch the **SPAN** switch on this unit.

Close release valve and, using hand pump and vernier adjustment, set the pressure until the "CURRENT" reading is 20mA on the display panel of the calibrator. The display on the transmitter should also indicate this reading.

As before, check and record the milliamp values of output for each 10% of pressure span. Plot these values to confirm linearity of the transducer and transmitter.

Set Pressure (kPa)	Required current (mA)	Measured current (mA)
0%	4	
10%	5.6	
20%	7.2	
30%	8.8	
40%	10.4	
50%	12.0	
60%	13.6	
70%	15.2	
80%	16.8	
90%	18.4	
100%	20.0	

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PART 2.

Zero the pressure in the system with the pressure release valve.

Rotate the pressure selection at the base of the hand pump in the anti-clockwise direction.

Connect the pressure calibrator to the **LOW** side of the Rosemount pressure transmitter using ¼" pipe.

Set a pressure of about 25 kPa.

What is the difference between the calibrator reading and the transmitter reading?

Refer to the Manufacturers brochure for the following:
What is the sensing device used? (Ref page 3)

What types of pressure can this device measure?
- Gauge
- Absolute
- Differential

What are the 3 types of process flanges, and what are they used for? (Ref page 8)

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Practical Exercise 2 Level

2.1 Objectives

- To demonstrate the use of ultrasonics in level measurement.
- To determine a suitable range of operation and provide a 4-20mA signal for that range.

2.2 Equipment Required

- Milltronics Ultrasonics probe mounted on tripod **LE 201**
- Milltronics Control panel and remote control unit **LT 201**
- Tape Measure **LI 202**

2.3 Ultrasonics configuration and familiarization

Often the requirement for level sensing may require that a non-contact and non-intrusive means may be required.

This may be due to a number of factors such as:

- temperature
- nature of the material
- distance

One of the most common types of non-contact sensors is the ultrasonic level measurement.

Procedure

Refer to the "Quick start instructions" for details on parameter entry.

When demonstrating this unit an audible recognition can be heard as the transducer "pings".

Apply power to the Milltronics Hydroranger ultrasonic level monitor, and check that the display is operational.

Angle the transducer vertically down towards the floor, (or towards a wall) with no objects between the transducer and the wall. This is intended to measure level by raising and lowering the tripod stem.

Note that when the tripod stem is fully retracted, then a high level is simulated. When the stem is fully extended, this is the low level. There is no need to adjust the tripod legs for this exercise.

Set the unit to measure level (Refer Step 1, P001), and consider the level to be a solid (Refer Step 2, P002).

The response time needs to be configured (although default settings work well in most applications). Note that setting a response quicker than that of the system can lead to spurious errors in measurement. A too slow response can make the system difficult to control

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in closed loop applications. Refer Step 3, P003. 10m/min will work well for our application since it is not critical control.

The transducer is a XPS-15 and should be set up with parameter no. 4 containing the value 104. Refer Step 4, P004.

Measurement units may vary. The SI Standard is metres and to maintain a high level of accuracy and engineering then milli-metres (mm) can be used.

Alarms trips and other operational limits also operate as a percentage of the range of operation.

Configure the operating range of the system as follows:

For this exercise, when the tripod stem is extended to its highest, this will be considered the low level or the tank's empty state.

Set the system to the empty state by ensuring that the tripod stem extension is wound down. (Please release the locks on the tripod first and do not use excessive force!) Note the distance using the measure. Enter this distance into parameter P006.

Retract the tripod stem to the **minimum** height. (Releasing the lock first!) This is considered the high level, or Tank Full state.

Note the difference between the high and low level readings (about 350mm?). This is the span or operating range of the system. Enter this distance into Parameter P007.

P001	1	Level – How full the vessel is
P002	2	Solid Process material
P003	3	Fast response –10m/min
P004	104	(Head Type: XPS-15)
P005	3	units of measurement - mm
P006	Low Level	Value
P007	Span	Value

2.4 Calibration of the 4-20mA output

Most ultrasonic units will monitor and indicate the level in the engineering units without much configuration required. However in a control system, or to other display devices the information can be sent by means of a 4-20mA signal. For this milliamp signal to be of use, the range needs to be configured to represent the level.

This unit can measure from 0.3m to 9.6m. If however the range of operation is quite small then the resolution and accuracy of the signal can be increased by reducing the range of the signal (4-20mA) to that of the system.

Refer to the users manual for the level sensor and configure the 4-20mA output for the range of operation as defined above.

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P200	2	Select 4-20mA output
P201	1	Level measurement
P202	1	

Remember:
- The low level occurs when the stem is fully extended and should show 0% - 4mA.
- The high level occurs when the stem is fully retracted and should show 100% - 20mA.

Check the operation as follows:
Pressing the [5/mA] key when in RUN mode will display the output current in the Auxillary reading window.

Vary the height of the transmitter and check the following:
Correct mA signal at low level
Correct mA signal at mid range
Correct mA signal at high level

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2.5 Ultrasonic functionality

Ultrasonic relies on the reflection of sound waves from the material being sensed.

Here we will look at the effectiveness and potential problems of sensing different materials.

Configure the sensor for DISTANCE measurement as follows:

P001	3	Distance – From the Transducer face
P002	2	Solid Process material
P003	3	Fast response –10m/min
P004	104	(Head Type: XPS-15)
P005	3	units of measurement - mm
P006	9000mm	maximum distance
P007	9000mm	maximum distance

TEST 1:

Place a Towel (or another sound absorbent media) in the path of the ultrasonic signal and check what distance is detected.

TEST 2:

Sense a solid object (wall) through a glass door or window. Which distance will be measured. Check target with light pen.

TEST 3:

Check the sensing target when the ultrasonic beam is reflected. The reflection can be off a table or glass window, onto a wall.
Start with a very shallow angle and determine that the wall is sensed. Use the light pen to check the target position.

Increase the angle until the sensor detect the reflecting media.

What is the angle of incidence?

Does this vary with different reflection media?

If so, why?

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Practical Exercise 3 Temperature

3.1 Objectives

- To calculate the response time for different thermocouple and RTD installations.
- To compare extension leads with compensation leads.

There are two sections to this practical exercise.

You may already find that the hardware connections have been made by a previous group. However please check **that they are correct.**

3.2 Equipment required:

- 1 x K Thermocouple **TT 301**
- 1 x J Thermocouple **TT 302**
- 2 x RTD - Pt 100 **TT 303, TT 304**
- Ice bath
- Boiling Water in container
- Datalogger and computer
- 2 x Thermowells
- Thermopaste
- Small Screwdriver suitable for screw terminals of Data Logger
- Thermocouple connected to both extension (**TT 305**) and compensation leads (**TT 306**)

3.3 Temperature response and accuracy

The response time is the time it takes to get to 63.2% of the final settling value. This is independent of the range of the device. If the device were a Pt 100 with a range of operation of -200 to 630°C and the temperature being monitored changed from 100 to 200 ^{o}C, then the response time would be the time it takes to go from 100 ^{o}C to 163.2 ^{o}C.

3.3.1 Procedure

Hardware Setup

1. Connect the 9-pin D-type male connector of the communications cable supplied to the data logger 9-pin female RS232 Communications socket.

2. Connect the other 9-pin female connnector to either the COM1 or COM2 port of the Personal Computer.

3. Power up the Personal Computer.

4. Connect the Plug pack power supply to the data logger AC/DC terminals. Although the output of the power supply is AC it doesn't matter which lead goes into which

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terminal as the terminals are polarity independent. (Be careful not to short circuit the Plug Pack terminals!)

5. Plug the power supply into the mains supply and power on the data logger. The LED on the logger should flash on and off.

6. Start the DeTerminal Program by changing to the DT subdirectory and typing DT <ENTER> at the DOS command prompt. The sequence of characters should be as follows:

   CD\DT <ENTER>
   DT <ENTER>

   If the data logger has been found connected correctly you will first get the user message:

   Found at 4800 baud

   The DeTerminal Program will then show a user screen, which has two windows. The top window displays data returned from the data logger and from now on will be referred to as the data window. The bottom window is used for typing commands to be sent to the data logger and will be referred to as the command window. Both these windows can be scrolled, saved and cleared.

Sending commands to the Data Logger

7. The currently highlighted line of information in the command window can be sent to the data logger by pressing the <ALT> and <L> keys together. This combination of keys is known as the Send Line command.
8. If you want to toggle between the screen windows, you can use the <TAB> key controls movement between the two screen windows. The active window can be identified by the options showing at the top border of the window.

Obtaining Readings

9. The best way to achieve this is by creating a whole series of readings which will be taken at regular intervals. To perform this task a schedule is sent to the logger. A schedule is a list which tells the device which channels to read and when.
10. A standard reporting schedule which reads the thermocouple temperature and voltage on channel 5 every second, can be sent to the data logger by sending the command line:

    RA1S 5TK 10V <ALT> <L>

    The meaning of the ASCII character sequence which makes up the schedule command is as follows:

- RA1S is the schedule command to report schedule A (or scan the channel list) every 1s.

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- 5TK is the channel command to read the K-type thermocouple temperature on channel 5.
- 10 V is the channel command to read the differential voltage input on channel 10.
- Additional channels may be added and read simultaneously e.g. RA1S 5TK 4TJ

11. To temporarily halt the schedule RA send the command line:

    HA <ALT> <L>

12. To continue send the command line:

    GA <ALT> <L>

Thermocouples

Connect the K-type sheathed thermocouple **TT 301** to channel 5 of the data logger as shown in the figure below.

Check equipment details:

Tag no.:	TT 301
Model:	K-type thermocouple
Details:	Sheath - Yellow
	Positive - Yellow
	Negative - Red

The ANSI MC96.1 standard for thermocouples and extension wire specifies that for 'K' type thermocouples the outer cover is yellow and the terminals are as follows:

- Positive (+) terminal is yellow.
- Negative (-) terminal is red.

Ensure that each lead is correctly positioned and securely tightened by the (+) and (-) screw terminals of channel 5.

The differential measurement configuration shown is most commonly used where the signals from the sensor/transducer are very small and more susceptible to the effects of noise and/or where the cables from the transducer to the measuring equipment are very long and also more susceptible to noise.

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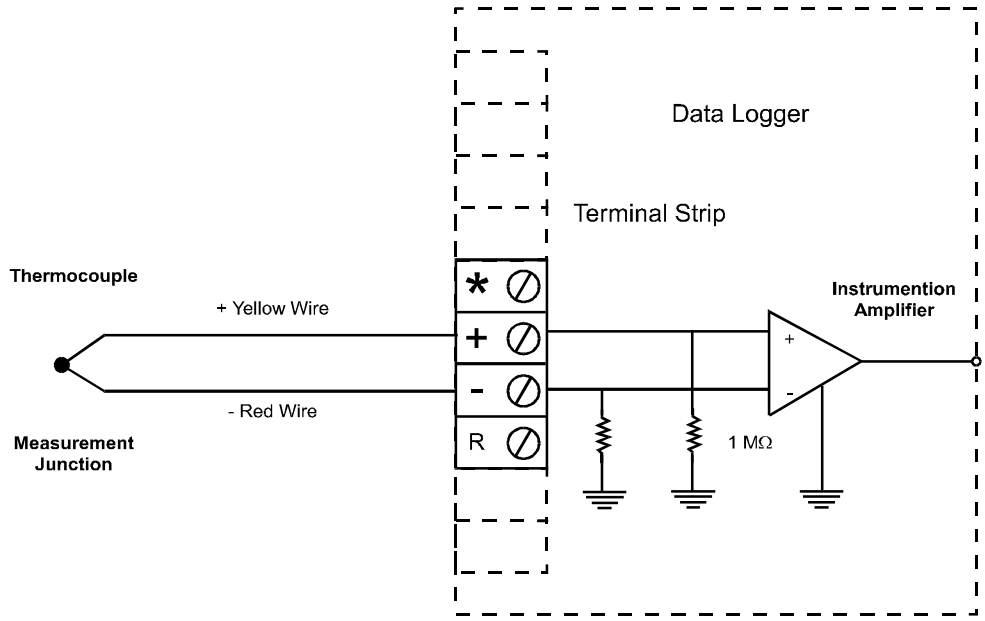


Figure E3.1
Differential Thermocouple Configuration.

In order to make a simultaneous continuous temperature measurement type the following at the highlighted command bar:

RA2S 5TK 4TJ <ALT><L>

Now connect in the J-type sheathed tip thermocouple to channel 4 using the same procedure as above. (The cover is black, the positive terminal white and the negative terminal is red).

Check equipment details:

Tag no.: TT 302
Model: J-type thermocouple
Details: Sheath - Black
Positive - White
Negative - Red

Now compare the response times for the two thermocouples in moving from ice to the hot water for the measurement to "settle down".

Examine the influence of thermowells on RTD response

Connect the RTD (without thermowell) to channel 1 of the data logger using the four-wire configuration as shown in the figure below.

Check equipment details:

Tag no.: TT 303
Model: RTD
Type: Pt100

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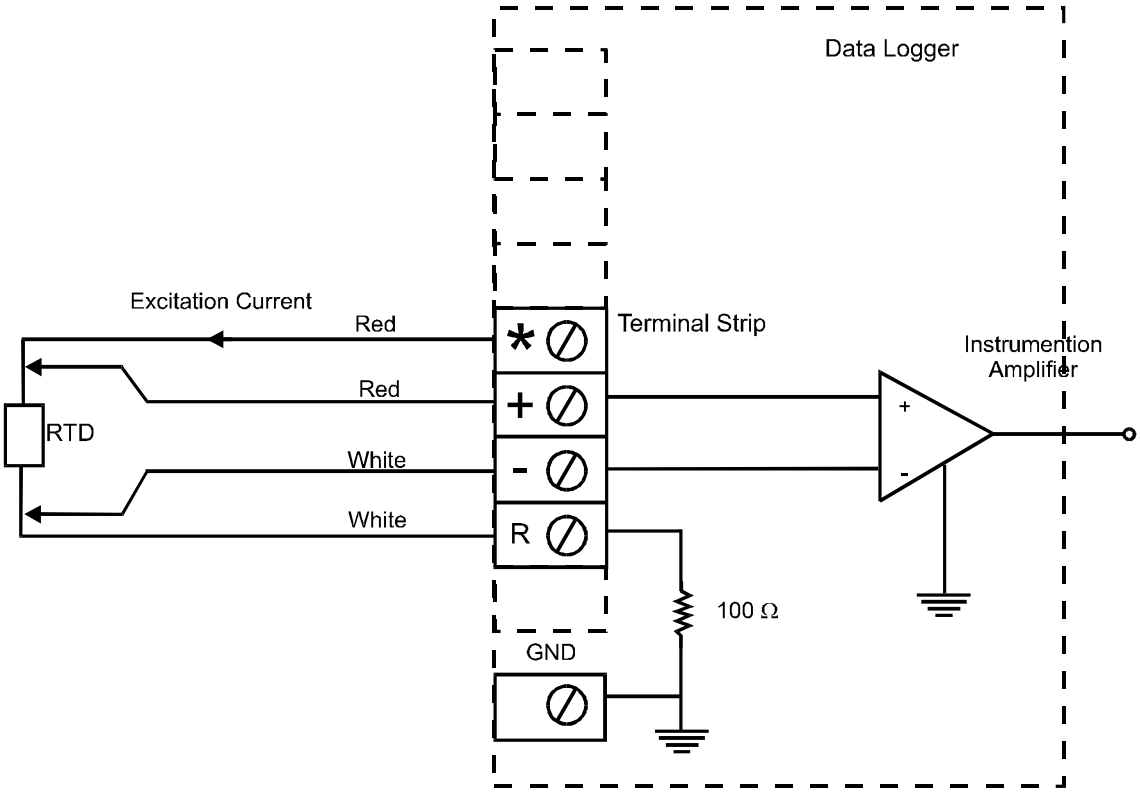


Figure E3.2
Four-wire RTD input.

In the four wire configuration two excitation leads carry the constant excitation current which flows through the RTD. The value of this current should remain constant irrespective of the excitation lead resistance or the resistance of the RTD.

The other two measurement leads are used to measure the voltage drop across the RTD. The actual current flowing in these leads is very small due to the very high input impedance of the voltage measurement circuit. Therefore, voltage drops across the lead resistances are so small they can be assumed to be negligible.

Now connect up channel 2 of the data logger with the second RTD.

Check equipment details:

Tag no.: TT 304
Model: RTD
Type: Pt100

Create a schedule of readings by typing:

RA2S 1PT385(4W) 2PT385(4W)

Take the RTD in a thermowell (but with no thermopaste) connected to channel 1 and see how long it takes to stabilise in being moved from the ice bath to the hot water. Repeat the exercise with the RTD in the thermowell connected to channel 2 but with thermopaste. (Be careful with the theropaste. It is very gluey!)

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3.4 Thermocouple extension leads compared with compensation leads

There are two basis means of increasing the distance for which a thermocouple can be located from a transmitter.

The two types of leads are:

- Extension leads
- Compensation leads

Extension leads are typically of the same material as the thermocouple junction, whereas compensation leads are copper or a copper alloy.

Procedure

Connect up a thermocouple which has both extension leads and compensation leads connected to it.

Check equipment details:

Tag no.: TT 305
Model: K-type thermocouple with extension leads
Details: Sheath - Yellow
Positive - Yellow
Negative - Red

Tag no.: TT 306
Model: K-type thermocouple with compensating leads
Details: Sheath - Red
Positive - Brown
Negative - Red

Identify which is a compensation lead and which is an extension lead. Confirm the type of thermocouple.

Using the procedure before, connect the extension leads to the channel 3 on the data logger and the compensation leads to channel 4 on the data logger.

Use the commands:

RA1S 3TK 4TK <ALT> <L>

Measure ice bath temperature and hot water with both set of leads and compare results.

What do you notice ?

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Practical Exercise 4 Simulation and Temperature Transmitter Calibration

4.1 Objective

- To demonstrate the use of signal or transmitter simulation.
- To demonstrate zero and span adjustment.
- To demonstrate conformance testing to specifications
- To demonstrate simulation of a temperature sensor (PT100)

Transmitter Simulation

Simulation of field equipment is often required for testing or pre-commissioning to check that the transmitter or other parts of the system are functioning correctly.

The most common signal for simulating a transmitter is 4-20mA and if often used to test the operation of a control alarm function before it is placed into service.

The basis of transmitter calibration

To ensure (and possibly certify) that a transmitter is functioning correctly and within specifications, two functions are required:

Simulation of the process measurement (ie. generation of a thermocouples millivolt signal)

Measurement of the transmitters output (ie. a 4-20mA meter)

More elaborate and expensive equipment can combine both of these functions, and simplify the test procedure.

In checking the accuracy, it is simply a procedure of setting the process input to the maximum and minimum values (and possibly a few points in between) and calculating the percentage error in the output from what is expected.

Calibration requires a little more effort. This requires the adjustment of the offset, commonly referred to as the zero, and also the adjustment of the range, also known as the span.

Zero and span adjustment

For accurate calibration the zero and span need to be set on the transmitter. The calibration of a transmitter is typically performed on the upper and lower range values. This requires the response to be linear. Checks are later performed on values mid range.

The zero adjustment typically sets the lower range limit, with the span used to set the upper range limit. However, adjusting the zero also moves the upper range valve, and adjusting the span can affect the lower range value, especially if it is not zero as in the case of 4-20mA transmitters.

The adjustment of zero and span is therefore alternated until acceptable values are achieved on the transmitter's output.

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4.2 Equipment Required

- Process calibrator (4-20mA signal generator - Fluke 743 Documentation Process Calibrator)
- Loop Powered Indicator (4-20mA) **EI 401**

4.3 Loop Power Simulation

4.3.1 Procedure

1. Connect "**mA source**" terminals (centre) from the calibrator to the Loop powered indicator terminals (I- and I+).
   Note :- Ensure DS and DPI jumpered. Why?

2. On the process calibrator, switch on with the **GREEN** power on button.

3. Next, press the **"MEAS/SOURCE"** button. This puts the process calibrator in the source screen.

4. Since we need to source milliamps, press **"mA"** once. Confirm "**Source mA**" with "**ENTER**".

5. The process calibrator will respond with ????? asking for a value to source. Key in **"4"** and **"ENTER"**, this will source 4mA. Confirm loop powered indicator displays a reading. Using "**OFFSET**" set reading to "**0.00**".

6. Change source to 20mA and press "**ENTER**".

7. Using "**SPAN**" control on loop powered indicator, set reading on display to 100.00.

8. Switch off process calibrator and disconnect loop powered indicator.

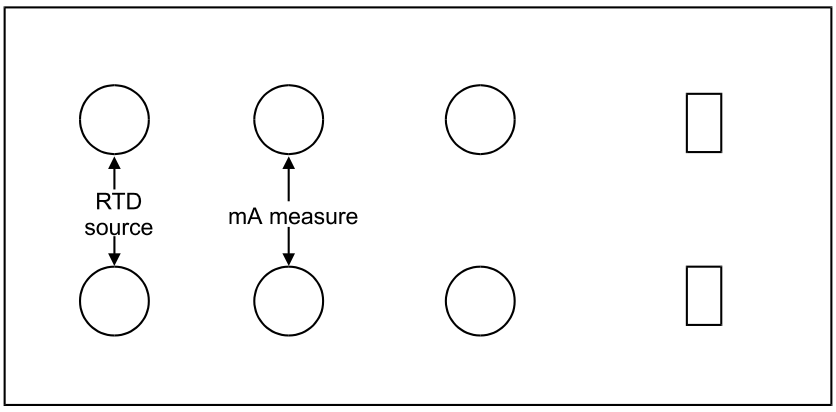


Figure E5.1

4.4 Equipment Required

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- Fluke 743 Documenting Process Calibrator
- Transmitter (RTD/4-20mA) **ET 402**

4.5 Zero and Span Adjustment

4.5.1 Procedure

1. Connect the "**RTD source**" terminals (left hand side) of process calibrator to the RTD input of the transmitter (Blue/White). This provides simulation of the process temperature. (Does it matter if connections are reversed?)

2. Connect the "**mA measure**" terminals (centre) of the process calibrator to the loop output of the transmitter (Red/Black). The process calibrator provides the loop power. This connection provides the measurement of the transmitters output signal (4-20mA). (What happens if these connections are reversed?)

3. On the process calibrator, switch on with the **GREEN** power on button. The default display is "**Measure mV**". Select the **"mA"** button to measure milliamps. Initially this will be zero.

4. Next, press the **"MEAS/SOURCE"** button. This puts the process calibrator in the source screen.

   As we need to source a RTD, press **"TC/RTD"** twice.

   Use the down arrow to move the cursor to **"PT100(385)"** and press **<ENTER>**.

   The process calibrator will respond with ????? asking for a temperature to source. Key in **<0>** and <**ENTER>**, this will source 0°C.

5. Use the **"MEAS/SOURCE"** button to bring up the split screen.

6. The next step is to configure the process calibrator to provide loop power.

   Press the **"SETUP"** button. The cursor should be on "**Loop Power (Disabled)".** Press **"ENTER**".

   Cursor down to the **ENABLE 24V** and press **"ENTER".**

   Confirm LED on transmitter comes "on". (What would be the cause if it fails to light?)

   Use the **"DONE"** blue key to confirm and exit.

   Note the 'Loop 24V' displayed at the top of the screen.

4.6 Calibrating a Transmitter

Now we will set transmitter to have a range of 0 to 100°C.

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1. With 0°C set on "**SOURCE**" screen adjust transmitter zero for 4mA on "**measure**" screen.

2. Next Select 100°C and adjust span for 20mA.

3. Return to 0°C and confirm holding of Cal – if not repeat 1 and 2 until values are reasonable.

4. Confirm linearity by selecting source of 25, 50, 75°C and checking the mA reading. (What should they be?)

5. Install RTD temperature probe and confirm values in ice water and ambient.

6. From the "**measure**" screen note the mA value and calculate the ambient temperature in °C.

7. As final confirmation of complete system operation connect the loop powered indicator into the 4-20mA loop (Figure E5.2). The reading on the display should confirm your calculated ambient temperature.

4.7 Error calculation using the Process Calibrator

From the split screen, select **"As Found"** and follow the prompts with the following configuration.

Select **"Instrument"** from top the list.

Enter the values as follows:

MEASURE	0% Value	4mA
	100% Value	20mA
	Tolerance	0.1%
	Delay	1s
SOURCE	2W RTD Pt100 (385)	
	0% Value	0°C
	100% Value	100°C
	Test Strategy	5^

Select the **"Done"** key when all values are entered.

Select the **"Auto Test"** to coduct the test on the transmitter.

The Process calibrator will conduct the test at 5 points, 0,25,50,75 and 100°C.

The summary of the errors will be shown in a tabulated form with errors exceeding the tolerance selected being shown in reverse video.

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Select the **"Done"** key when finished viewing, and the Process Calibrator opens data entry windows for Tag and serial number information.

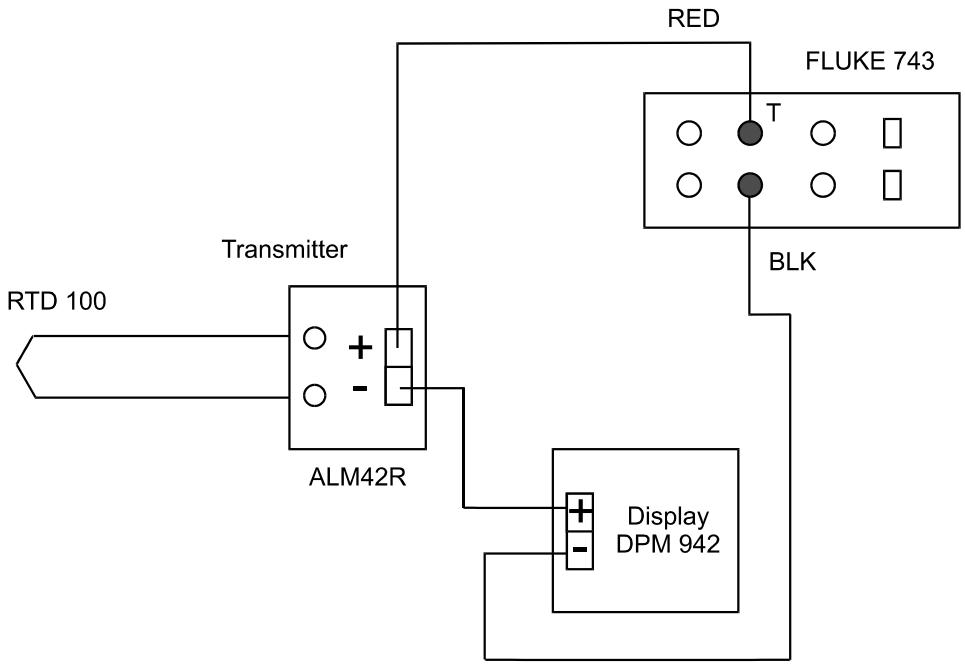


Figure E5.2

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Practical Exercise 5 Valve sizing (software)

5.1 Objectives

- To illustrate the features expected of sizing software for valves

A typical vendor's package is used to demonstrate the features available from a number of valve sizing software packages. The packages available are normally targetted a particular vendor's products are not always available for general use (except by the vendor's staff).

5.2 Equipment Required

- Personal Computer
- Software loaded in directory C:\Keystone

5.3 Procedure

1. Type:
   CD\KEYSTONE <ENTER>
   SIZING <ENTER>

2. In answer to the query:

   Select sizing program for Liquids.
   By typing :

   1 <ENTER>

3. Select Flow Rate as the next screen.
   By typing:
   3<ENTER>

4. Select Metric Litres/sec
   3 <ENTER>
5. Select Water as the medium.
   W <ENTER>
6. Select Water
   1 <ENTER>
7. Select General Purpose F1/F2
   1 <ENTER>

8. The remaining parameter that need to be entered are:
   Valve Size: 50 mm
   Valve Angle: 45 degrees
   Upstream Pressure: 350 kPa
   Downstream Pressure: 250 kPa
   Diameter downstream: 50 mm
   Diameter upstream: 50 mm

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9. Perform the calculation and find out where cavitation occurs. Once you have done this recalculate for different values of upstream and downstream pipe diameter. What do you notice ?

Cavitation is reported as two different types in this program:

Incipient where cavitation is just starting.
Critical where the valve is fully choked.

THREE EXERCISES IN VALVE SIZING

Introduction:
Quite often a valve is "sized" according to the dimensions of the end connections, whereas in reality a valve "size" is the flow the valve can provide, and this can be determined by using readily available data. If a valve offers very little restriction to flow, above and beyond the flow control element housed within it, the sizing factor is better than one where other flow restrictions, such as varying diameters and change of flow direction through the valve are present.

The principle of flow calculation, particularly in liquids which have minimal change in volume/pressure ratios, is relatively simple. First, if we consider a fixed orifice plate flowmeter, where we have the following factors present:

1. Upstream or inlet pipe diameter
2. Upstream or inlet pressure
3. A Restriction (Orifice Plate) that has two factors,
   Orifice diameter (the hole in the plate, which is always smaller than the pipe diameter)
   Orifice hole shape
4. The downstream or outlet pressure
5. Fluid density

With this information we can, by comparing the pressure difference between the inlet and outlet, calculate a flow rate.

The same applies for valves, to size one we need to know the differential (Inlet to Outlet pressure difference) pressure, and the fluid density.
Additionally we need to know the characteristics of the valve,. As explained above, these are things like changing diameters through the valve and changes in fluid flow direction.

These fairly complex factors are merged together, and the result of this algorithm is known as the "Valve Flow Coefficient" or "C_V" which is obtainable from the valve manufacturer.

This results in the following equation $q = N1Cv\sqrt{\frac{\Delta p}{G1}}$

Where:

q = Flow Rate
N1 = Unit Constant
U.S. Gallons/Min = 1,
Imperial Gallons/Min = 1.16.
Litres/Min = 14.42 etc.)

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Cv = Valve coefficient
Δ p = Pressure differential
G1 = Viscosity

Translating this equation into English, we get

$$\text{Flow Rate} = \text{Unit constant} \times \text{Valve Characteristic} \times \sqrt{\frac{(\text{Inlet pressure} - \text{Outlet Pressure})}{\text{Specific Gravity of fluid.} \quad (\text{Water} = 1)}}$$

E X E R C I S E S

1). Differences caused by the Specific Gravity. Note the relationship between Flow & Sg

- A 25 mm valve has a Cv of 1.68
- An inlet pressure of 600 kPa
- An Outlet pressure of 515 kPa

Calculate the flow rates in Litres/min. of the following fluids

a). Water Sg = 1.0
b). Sulfuric Acid Sg = 1.8 (The internal table may be missing for this! If so please ignore)
c). Ether Sg = 0.74

2). Valve Rangeability

The same 25 mm valve has, at the listed % opening, the following pressure differentials:

% of Valve Opening	Pressure Differential kPa	Calculated Flow Rate
10%	400	
20%	397	
30%	392	
40%	384	
50%	371	
60%	350	
70%	316	
80%	261	
90%	172	
100%	28	

Calculate the flow rates obtainable through it

- Assume a Sg = 1.05
- A Supply pressure of 1200 kPa

3). (Q) What type of characteristic does this valve display?

4). It is required to control the flow of a liquid with a Sg of 1.25 @ 20^0C from a source of 3500 kPa and a controlled flow rate of between 20 and 45 Litres/Sec.

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The maximum allowable pressure differential is 350 kPa @ 20 litres/sec.

What is the minimum size valve that can be used ? (Assume a Cv of 3.45)

Remember

- *Cv is the number of American gallons of water that can pass through a valve in one minute with a pressure drop of 1 PSI at ambient temperature (PSI meaning Pounds per Inch ^{2})*
- *1 PSI = 6.895 kPa*
- *1 US Gallon = 3.7854 Litres*
- *60 Seconds = 1 minute*

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